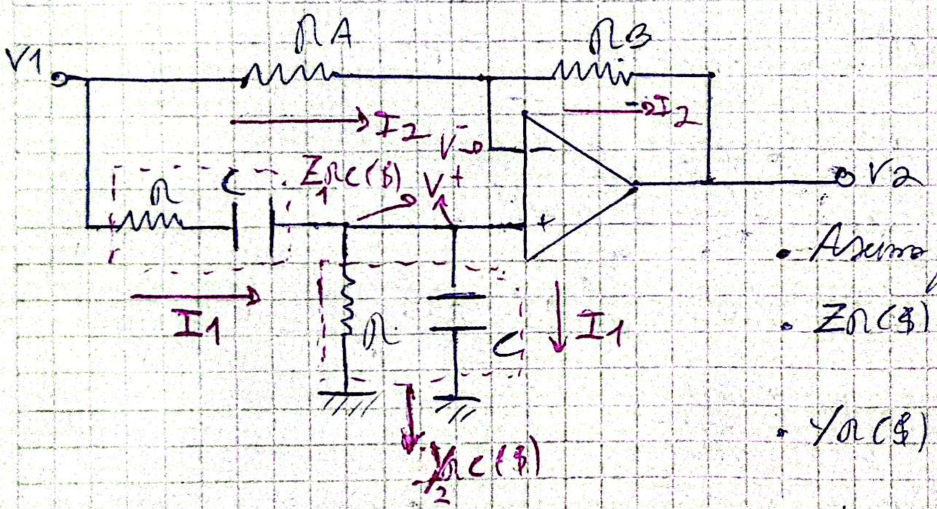
Ejercicio 4)

b)

• Filtro Pasa Todo o Rotador de Fase, queremos $H(s) = \frac{V_2(s)}{V_1(s)}$ tanto el módulo, la fase, y el diagrama de polos y ceros

• Tomo $\frac{R_2}{R_1} = 1$ y $\frac{R_A}{R_B} = 5$



• Asumo filtro ideal

• $Z_R(s) = R$, $Z_C(s) = \frac{1}{sC}$

• $Y_R(s) = \frac{1}{R}$, $Y_C(s) = sC$

• $V^+ = V^-$ por equipotencialidad

• $V^+ = V^- = V_1(s)\,\dfrac{Y_{1RC}(s)}{Y_{1RC}(s) + Y_{2RC}(s)}$, $I_{in}(s) = I_1(s) + I_2(s)$

• $I_1(s) = (V_1(s) - V^+)\cdot Y_{1RC}(s) = (V^+ - 0V)\cdot Y_{2RC}(s)$ → Por igualación de corrientes y análogamente al ejercicio 4) a) => se obtiene

$V^+ = V_1(s)\,\dfrac{Y_{1RC}(s)}{Y_{1RC}(s) + Y_{2RC}(s)}$ con $V^+ = V^-$

• $I_2(s) = (V_1(s) - V^-)\cdot Y_{RA}(s) = (V^- - V_2(s))\cdot Y_{RB}(s)$

$\Rightarrow V_1(s)\cdot Y_{RA}(s) - V^-\cdot Y_{RA}(s) = V^-\cdot Y_{RB}(s) - V_2(s)\cdot Y_{RB}(s)$

$\Rightarrow V_1(s)\cdot Y_{RA}(s) + V_2(s)\cdot Y_{RB}(s) = V^-\cdot Y_{RB}(s) + V^-\cdot Y_{RA}(s)$

$V_1(s)\cdot Y_{RA}(s) + V_2(s)\cdot Y_{RB}(s) = V^-\cdot\left(Y_{RB}(s) + Y_{RA}(s)\right)$

$$V_1(s)\cdot Y_{RA}(s) + V_2(s)\cdot Y_{RB}(s) = V_1(s)\cdot \frac{Y_{1RC}(s)}{Y_{1RC}(s)+Y_{2RC}(s)}\cdot\left(Y_{RB}(s)+Y_{RA}(s)\right)$$

$$V_2(s)\cdot Y_{RB}(s) = V_1(s)\left(\frac{Y_{1RC}(s)\cdot Y_{RB}(s)}{Y_{1RC}(s)+Y_{2RC}(s)} + \frac{Y_{1RC}(s)\cdot Y_{RA}(s)}{Y_{1RC}(s)+Y_{2RC}(s)}\right) - V_1(s)\cdot Y_{RA}(s)$$

$$\frac{V_2(s)}{V_1(s)} = \frac{Y_{1RC}(s)\cdot \cancel{Y_{RB}(s)}}{Y_{1RC}(s)+Y_{2RC}(s)}\cdot\frac{1}{\cancel{Y_{RB}(s)}} + \frac{Y_{1RC}(s)\cdot Y_{RA}(s)}{Y_{1RC}(s)+Y_{2RC}(s)}\cdot\frac{1}{Y_{RB}(s)} - \frac{Y_{RA}(s)}{Y_{RB}(s)}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{Y_{1RC}(s)}{Y_{1RC}(s)+Y_{2RC}(s)}\left(1+\frac{Y_{RA}(s)}{Y_{RB}(s)}\right) - \frac{Y_{RA}(s)}{Y_{RB}(s)}$$

- $$Y_{1RC}(s) = \frac{1}{Z_{1RC}(s)} = \frac{1}{R+\frac{1}{sC}}\cdot\frac{sC}{sC} = \frac{sC}{sCR+1}$$

- $$Y_{2RC}(s) = \frac{1}{Z_{2RC}(s)} = Y_{2R}(s)+Y_{2C}(s) = \frac{1}{R}+sC = \frac{1+sCR}{R} = \frac{sCR+1}{R}$$

- $$Y_{RA}(s) = \frac{1}{R_A},\quad Y_{RB}(s) = \frac{1}{R_B}$$

- $$\frac{Y_{1RC}(s)}{Y_{1RC}(s)+Y_{2RC}(s)} = \frac{\frac{sC}{sCR+1}}{\frac{sC}{sCR+1}+\frac{sCR+1}{R}} = \frac{sC}{(sCR+1)\left[\frac{sC}{sCR+1}+\frac{sCR+1}{R}\right]}$$

$$= \frac{sC}{sC+\frac{(sCR+1)^2}{R}}\cdot\frac{R}{R} = \frac{sCR}{sCR+(sCR+1)^2} = \frac{sCR}{sCR+s^2(CR)^2+2sCR+1}$$

$$= \frac{sCR}{s^2(CR)^2+3sCR+1}\cdot\frac{1/(CR)^2}{1/(CR)^2} = \boxed{\frac{s/CR}{s^2+3s\frac{1}{CR}+\frac{1}{(CR)^2}} = \frac{Y_{1RC}(s)}{Y_{1RC}(s)+Y_{2RC}(s)}}$$

- $$\boxed{\frac{Y_{RA}(s)}{Y_{RB}(s)} = \frac{1/R_A}{1/R_B} = \frac{R_B}{R_A}}$$

Escaneado con CamScanner

$$\boxed{H(s)=\frac{s/CR}{s^2+3s\frac{1}{CR}+\frac{1}{(CR)^2}}\left(1+\frac{R_B}{R_A}\right)-\frac{R_B}{R_A}} \quad (4)$$

$$H(s)=\frac{\frac{s}{CR}\cdot R_A\left(1+\frac{R_B}{R_A}\right)-R_B\cdot\left(s^2+s\frac{3}{CR}+\frac{1}{(CR)^2}\right)}{\left(s^2+s\frac{3}{C\cdot R}+\frac{1}{(CR)^2}\right)R_A}$$

$$H(s)=\frac{s\cdot\frac{R_A}{C\cdot R}+\frac{s\cdot R_B}{C\cdot R}-s^2R_B-s\cdot\frac{3R_B}{CR}-\frac{R_B}{(CR)^2}}{s^2R_A+s\cdot\frac{3\cdot R_A}{C\cdot R}+\frac{R_A}{(C\cdot R)^2}}$$

$$\boxed{H(s)=\frac{-s^2R_B-s\cdot\frac{2R_B}{C\cdot R}+s\cdot\frac{R_A}{CR}-\frac{R_B}{(CR)^2}}{s^2R_A+s\cdot\frac{3\cdot R_A}{C\cdot R}+\frac{R_A}{(C\cdot R)^2}}} \rightarrow \text{Transferencia, expresió general}$$

• Ahora aplico la relación $\frac{R_A}{R_B}=5 \Rightarrow \frac{R_B}{R_A}=\frac{1}{5}$

$\Rightarrow$ retomando (4) $\Rightarrow H(s)=\dfrac{\frac{s}{C\cdot R}}{s^2+3\frac{s}{CR}+\frac{1}{(CR)^2}}\left(1+\frac{1}{5}\right)-\frac{1}{5}$

$$H(s)=\frac{\frac{s}{CR}\cdot\frac{6}{\cancel{5}}\cdot\cancel{5}-\left(s^2+3\frac{s}{CR}+\frac{1}{(CR)^2}\right)}{\left(s^2+3\frac{s}{CR}+\frac{1}{(CR)^2}\right)\cdot 5}$$

$$H(s)=\frac{-s^2-3\frac{s}{CR}+6\frac{s}{CR}-\frac{1}{(CR)^2}}{s^2+3\cdot\frac{s}{C\cdot R}+\frac{1}{(CR)^2}}\cdot\frac{1}{5}, \quad K=\frac{1}{5}$$

$$\boxed{H(s)=\frac{-s^2+3\frac{s}{C\cdot R}-\frac{1}{(C\cdot R)^2}}{s^2+3\frac{s}{C\cdot R}+\frac{1}{(CR)^2}}\cdot\frac{1}{5}} \rightarrow \text{con } \frac{R_A}{R_B}=5$$

NOTA

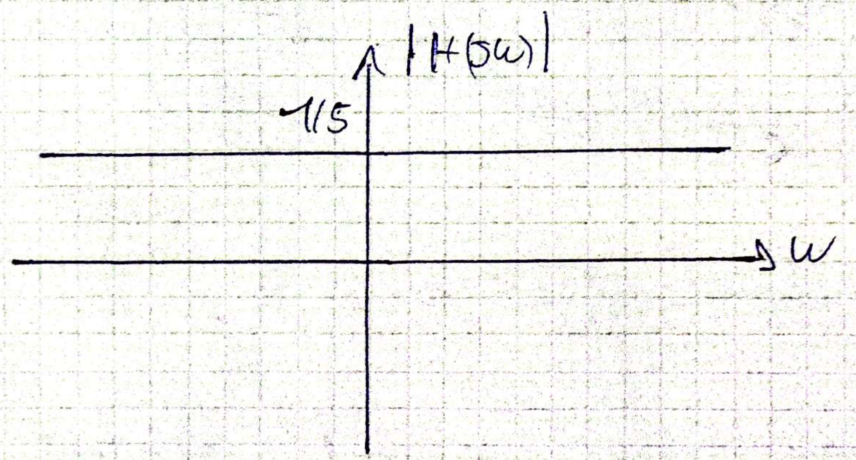
Busco el módulo

$$s = jw \Rightarrow H(jw) = \frac{+(jw)^2 - 3\cdot\frac{jw}{C\Omega} + \frac{1}{(C\cdot\Omega)^2}}{(jw)^2 + 3\cdot\frac{jw}{C\Omega} + \frac{1}{(C\Omega)^2}} \cdot \frac{1}{5}\cdot(-1)$$

$$H(jw) = \frac{-w^2 - j\cdot\frac{3w}{C\cdot\Omega} + \frac{1}{(C\cdot\Omega)^2}}{-w^2 + j\frac{3w}{C\Omega} + \frac{1}{(C\Omega)^2}} \cdot \frac{1}{5}$$

$$|H(jw)| = \frac{1}{5}\,\frac{\sqrt{\left(+\frac{1}{(C\Omega)^2} - w^2\right)^2 + \left(-\frac{3w}{C\Omega}\right)^2}}{\sqrt{\left(+\frac{1}{(C\Omega)^2} - w^2\right)^2 + \left(\frac{3w}{C\Omega}\right)^2}} = \boxed{\frac{1}{5} = |H(jw)|}$$

$\boxed{\bullet \text{ Paso todo y me atenúa } \frac{1}{5}}$



Busco la fase:

$$\varphi_H(w) = \operatorname{arctg}\left(\frac{Im\{Num\ H(jw)\}}{Re\{Num\ H(jw)\}}\right) - \operatorname{arctg}\left(\frac{Im\{Den\ H(jw)\}}{Re\{Den\ H(jw)\}}\right)$$

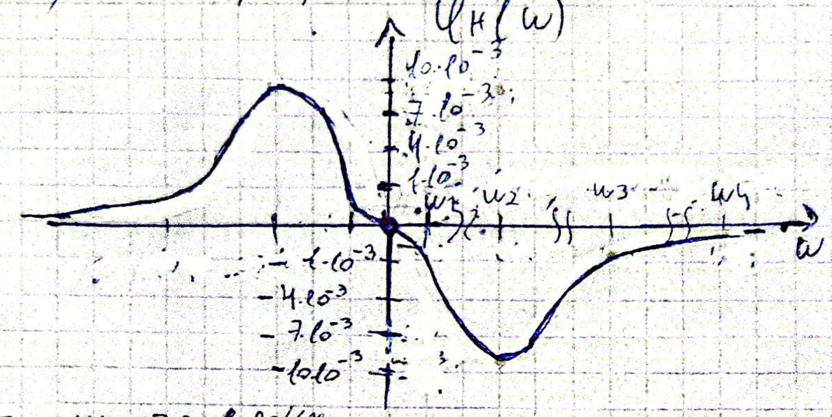
$$\varphi_H(w) = \operatorname{arctg}\left(\frac{-\frac{3w}{C\cdot\Omega}}{\frac{1}{(C\cdot\Omega)^2} - w^2}\right) - \operatorname{arctg}\left(\frac{\frac{3w}{C\cdot\Omega}}{\frac{1}{(C\cdot\Omega)^2} - w^2}\right)$$

$$\varphi_H(w) = -\operatorname{arctg}\left(\frac{\frac{3w}{C\cdot\Omega}}{\frac{1}{(C\Omega)^2} - w^2}\right) - \operatorname{arctg}\left(\frac{\frac{3w}{C\cdot\Omega}}{\frac{1}{(C\Omega)^2} - w^2}\right)$$

$$\varphi_H(w) = -2\operatorname{arctg}\left(\frac{\frac{3w}{C\cdot\Omega}}{\frac{1}{(C\Omega)^2} - w^2}\right)$$

$$\varphi_H(w) = -2\,\text{arctg}\left(\frac{\frac{3w}{C\cdot R}}{\frac{1}{(CR)^2} - w^2}\right) = -2\,\text{arctg}\left(\frac{3w}{-w^2 CR + \frac{1}{(CR)}}\right)$$

$C\cdot R = 10^{-3}$, con $R = 1k\Omega$, $C = 1\mu F$



- $w = 0 \Rightarrow \varphi_H(0) = 0$
- $w \to +\infty \Rightarrow \varphi_H(w) \to 0$
- $w \to -\infty \Rightarrow \varphi_H(w) \to 0$

- $f_1 = 10\,Hz \Rightarrow w_1 = 2\pi\cdot 10$ rad/seg
- $f_2 = 100\,kHz \Rightarrow w_2 = 2\pi\cdot 100\cdot 10^3$ rad/seg
- $f_3 = 1\,MHz \Rightarrow w_3 = 2\pi\cdot 1\cdot 10^6$ rad/seg
- $f_4 = 10\,MHz \Rightarrow w_4 = 2\pi\cdot 10\cdot 10^6$ rad/seg

$\varphi_H(w)$ rad	f
	0
$\equiv \mp 0{,}37\cdot10^{-3}$	± 10 Hz
$\equiv \mp 4{,}5\cdot10^{-3}$	± 100 KHz
$\equiv \mp 0{,}95\cdot10^{-3}$	± 1 MHz
$\equiv \mp 0{,}095\cdot10^{-3}$	± 10 MHz

Diagrama de polos y ceros

$$H(s) = \frac{-s^2 + 3\frac{s}{C\cdot R} - \frac{1}{(C\cdot R)^2}}{s^2 + 3\frac{s}{C\cdot R} + \frac{1}{(C\cdot R)^2}} \cdot \frac{1}{5}$$

Si $C\cdot R = 10^{-3}$ con $R = 1k\Omega$, $C = 1\mu F$

$$H(s) = \frac{(s - 2618{,}03)\cdot(s - 381{,}966)}{(s + 2618{,}03)\cdot(s + 381{,}96)} \cdot \frac{1}{5}$$

- Numerador da "ceros" $\to s_{01} = 2618{,}03$, $\to s_{02} = 381{,}96$
- Denominador da "polos" $\to s_{P1} = -2618{,}03$, $\to s_{P2} = -381{,}96$